SOFTWARE REQUIREMENT SPECIFICATIONS

OPTICAL HEART RATE DETECTION using WEBCAM

Version: 1

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Table of Contents

1. INTRODUCTION 3

1.1 PURPOSE 3

1.2 INTENDED AUDIENCE 3

1.3 PROJECT SCOPE 3

1.4 OPERATING ENVIRONMENT 4

1.5 DESIGN AND IMPLEMENTATION CONSTRAINTS 4

2. SYSTEM FEATURES 5

2.1 SYSTEM FEATURES 5

3. EXTERNAL INTERFACE REQUIREMENTS 6

3.1 USER INTERFACES 6

3.2 HARDWARE INTERFACES 6

3.3 SOFTWARE INTERFACES 6

3.4 COMMUNICATIONS INTERFACES ..... **Error! Bookmark not defined.**

4. OTHER NONFUNCTIONAL REQUIREMENTS 7

5. REFERENCES 7

1. INTRODUCTION

1.1 PURPOSE

Heart Rate (HR) is one of the most important Physiological parameters and a vital indicator of people's physiological state and is therefore important to monitor. Monitoring of HR often involves high costs and complex application of sensors and sensor systems. Research progressing during last decade focuses more on noncontact based systems which are simple, low-cost and comfortable to use. Still, most of the noncontact based systems are fit for lab environments in offline situation but needs to progress considerably before they can be applied in real time applications.

This paper presents a real time HR monitoring method using a webcam of a laptop computer. The heart rate is obtained through facial skin color variation caused by blood circulation. Three different signal processing methods such as Fast Fourier Transform (FFT), Independent Component Analysis (ICA) and Principal Component Analysis (PCA) have been applied on the color channels in video recordings and the blood volume pulse (BVP) is extracted from the facial regions. HR is subsequently quantified and compared to corresponding reference measurements. The obtained results show that there is a high degree of agreement between the proposed experiments and reference measurements.

This technology has significant potential for advancing personal health care and telemedicine. Further improvements of the proposed algorithm considering environmental illumination and movement can be very useful in many real time applications such as driver monitoring

1.2 INTENDED AUDIENCE

While the software requirement specification (SRS) document is written for a more general audience, this document is intended for individuals directly involved in the development of this project. This includes software developers, project consultants, team managers, users, testers, and documentation writers.

1.3 PROJECT SCOPE

Cardiac pulse is typically measured in clinical settings using electrocardiogram (ECG), which requires patients to wear chest straps with adhesive gel patches that can be abrasive and become uncomfortable for the user. Heart rate may also be monitored using pulse oximetry sensors that may be worn on the fingertip or earlobe. These sensors are not convenient for long-term wear and the pressure can become uncomfortable over time. In addition to the discomforts of traditional pulse measurement devices, these devices can damage the fragile skin of premature newborns or elderly people.
For these populations especially, a non-contact means of detecting pulse could be very beneficial. Non-contact heart rate measurement through a simple webcam or phone camera would also aid telemedicine and allow the average person to track their heart rate without purchasing special equipment. As part of the recent gain in popularity of fitness apps and the quantified self, regular non-obtrusive monitoring through a computer or phone camera may help detect changes in a person's heart

rate over time and indicate changing health or fitness. Heart rate can be detected without contact through photo-plethysmography (PPG), which measures variations in blood volume by detecting changes in light reflectance or transmission throughout the cardiovascular pulse cycle. PPG is usually performed with dedicated light sources with red or infrared wavelengths, as is the case for pulse oximetry sensors. Plethysmographic signal could also be detected in video from a regular color camera. They found that the signal could be detected within the red, green, and blue channels of color video of exposed skin, but that it was strongest in the green channel, which corresponds to the fact that hemoglobin has absorption peaks for green and yellow light wavelengths. They also found that although the signal could be detected in multiple locations on the body, it was strongest on the face, especially on the forehead. Although the plethysmographic signal may be detected in the raw color channel data, it is mixed in with other sources of color variation such as changes in ambient light or motion. The signal could be better extracted by using independent component analysis (ICA) to separate independent source signals from the mixed color signals

ALGORITHM
(1) Perform face detection on one frame of the video
(2) Decompose this frame into sub frames using the Gaussian pyramid decomposition algorithm. Thus the highest level of the pyramid has low pass information.
(3) Filter each level of the pyramid using a Band Pass Filter, with cut-off frequency given as [(hifreq*frame rate/sampling rate)±1] .
(4) Magnify the intensities of the frequency band of interest by a magnification factor.
(5) Collapse the pyramid into a single frame.
(6) Repeat the above 5 steps for all the remaining frames of the video.

1.4 OPERATING ENVIRONMENT

The software is implemented on a Windows environment using Python progamming language aided by three libraries: OpenCV, Numpy and Scipy.
The OpenCV library provides the necessary resources to the facial detection algorithm and the frame capture through the webcam connected to a computer. Numpy and Scipy are used for optimization, stats and signal processing.
Although it was not tested here, the software developed does not rely on any OS specific functionality, and thus can be ported to a Linux environment

1.5 DESIGN AND IMPLEMENTATION CONSTRAINTS

PPG is known to be susceptive to motion-induced signal corruption and overcoming motion artifacts presents one of the most challenging problems.
A good Signal-to-Noise Ratio (SNR) is essential to accurately detect the heart rate signal. It can be improved by either reducing the noise or increasing the signal power. In our case, the noise affecting the measurement comes from three sources basically:

- Image noise: It originates in the camera sensor. By averaging all the pixels for the brightness calculation, we are filtering out a big part of its spatial component. The band-pass filter that we are applying to the brightness signal is also rejecting much of its time component. However, there will always be some level of this noise in the band of interest.

- User behavior: The pressure variations of the fingertip against the camera lens may show up in

the signal. If the user has shaky hands, it is better that she uses a finger of the hand holding the phone.

- Lighting changes: Any change in the light sources or in the scene reflecting that light towards the camera may introduce noise in our pass band. For instance, moving the phone in the air while recording the video sequence may introduce artifacts if the scene parts in the field of view have very different light intensities.

2. SYSTEM FEATURES

2.1 SYSTEM FEATURES

The main features of the proposed method is 3 independent signals which are called Red signals, Green signals and Blue signals and these signals were produced from the red, blue and green color values of each pixel of all the facial image frames.

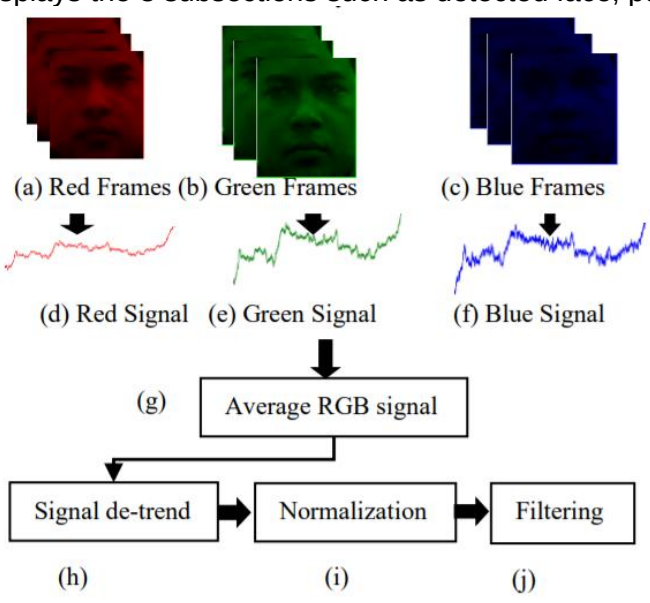
#	Title of System Feature	Description	Priority	Functional Requirements
1	*Frame capture*	*The first stage consists of the acquisition of a frame through a webcam connected to the computer. As it is usual with most webcams, the captured frame is in the RGB colour format*	*High*	*Although the resolution of webcam is not so important (a webcam with a 1280 × 720 resolution and 30 fps is used in the experiments), it is paramount that the webcam is able to capture the raw data, without any compression, as this would result in the loss of the color variations we are seeking. This is usually not a problem for webcams that are directly connected to the computer, but it must be observed for webcams connected through any kind of wired or wireless networks.*
2	*Face Detection and ROI Definition*	*The second stage consists of the facial detection over the acquired frame. As default, only one face is to be detected in this project, ignoring any other subjects appearing in the image. The third stage consists of defining a bounding box over the previously detected face.*	*Medium*	*The ROI is defined as h × w, where h is the height of the bounding box of the detected face and w is 55% of the width of this bounding box. The position of the ROI is centralized within the detected face The face detection algorithm is only performed for the first frame in each second (i.e., once every 30 frames). The ROI is kept the same for the remaining frames.*
3	*Trace Extraction*	*In the fourth stage, for every frame n, the ROI consists of a color image h×w with three channels, Red, Green and Blue.*	*High*	*The image is normalized, removing the mean value for each channel, making the variance between samples unitary as follows:* $Cbi(n) = Ci(n) - \mu i \sigma i$ *where i is for R, G and B, μi and σi are the mean and standard deviation for each channel, respectively.*
4	*Frequency*	*In the fifth stage, the signal of*	*High*	*The algorithm takes the trace values for the last 10*

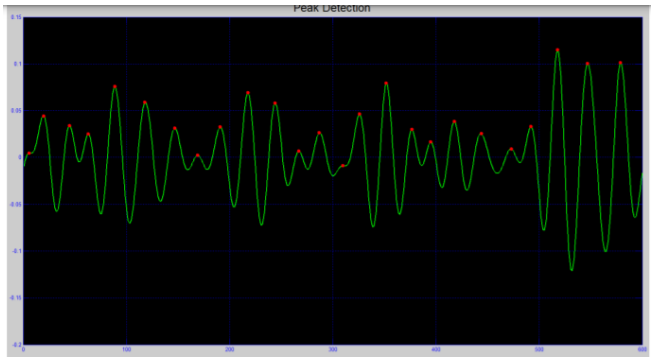
	Domain Filtering:	*interest is the trace, T (n). These algorithms are not applied for every new trace sample (i.e., every frame), but rather once every two seconds (i.e., once every 60 frames).*		*seconds and store it in a vector. It is then transformed to the frequency domain with a Fast Fourier Transform - FFT of 2048 points.*
5	*Peak Detection and Recurring Point of Interest Analysis*	*This is the sixth stage, which performs the heart rate estimation by analyzing the existing peaks in the obtained signal after the previous stage.*	*High*	*It is triggered only when the FFT is performed (i.e., every 2 seconds). The analyzed range is between 31 and 185 bpm, which corresponds to [0.52 3.10] Hz. The five magnitude peaks have their X-axis values (the bpm) stored and are considered as points of interest*

3. EXTERNAL INTERFACE REQUIREMENTS

3.1 USER INTERFACES

A Graphical User Interface (GUI) has been developed using MATLAB to monitor HR in real time which has 3 main sub-sections. Fig. shows the real time HR monitoring Graphical User Interface (GUI) which displays the 3 subsections such as detected face, pulse peak and current HR using the three methods.





3.2 HARDWARE INTERFACES

A webcam or a laptop's built-in camera is the only hardware requirement

3.3 SOFTWARE INTERFACES

Windows OS was used for its user friendliness but the project could easily be transferred to Linux based OS. The project uses various python libraries.

4. OTHER NONFUNCTIONAL REQUIREMENTS

#	Type of Requirement	Details
1	Performance Requirements	*Since the software works in real-time, the capture can take about a minute.*
2	Scalability Requirements	*A single person should be in the frame for capture. The software has not been scaled.*
3	Security Requirements	*The software does not require any user identity authentication or Network IP requirements since no data is collected or stored.*

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